

# Meher Chaitanya Pindiprolu, PhD

MACHINE LEARNING RESEARCHER | Efficient AI | LLM Optimization | Graph Machine Learning

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## Profile

Machine Learning Researcher with a PhD in Network Science from ETH Zürich, specializing in compute- and memory-efficient AI. Leads research on LLM pretraining efficiency, Tokenized Graph Transformers, long-range information flow, and scalable graph machine learning using optimization and numerical linear algebra. Combines research leadership with experience translating novel algorithms into production AI systems with measurable impact.

## Core Expertise & Skills

### AI:

Efficient LLM Training | KV-Cache Optimization | Graph Machine Learning | Long-Range Information Flow | Graph Transformers | Graph Neural Networks | Recommender Systems

### Methods:

Numerical Linear Algebra | Optimization | Spectral Methods | Graph Algorithms | Randomized Algorithms

### Programming & Frameworks:

Python | C++ | C | PyTorch | PyTorch Geometric | DGL | scikit-learn | NetworkX

## Work Experience

### KTH Royal Institute of Technology

Stockholm, Sweden

WALLENBERG AI POSTDOCTORAL RESEARCHER

2025– Present

- Define and lead research on efficient AI, originating projects in LLM pretraining efficiency, tokenized graph transformers, long-range information flow, and numerical methods for scalable graph learning.
- Drive end-to-end research from problem formulation to publication, including sparse Graph Transformers with up to 11% gains and large-scale LLM optimization experiments.
- Provide scientific direction to PhD and Master's researchers, shaping methodology, experimental design, interpretation, and cross-university collaborations.

### Coverstar

Stockholm, Sweden

AI CONSULTANT – RECOMMENDER SYSTEM ARCHITECT (part-time)

2025-2026

- Architected Coverstar's next-generation "For You" recommender, combining graph retrieval, session Transformers, cold-start methods, LLM-assisted content understanding, ranking, diversity, and safety.
- Identified that short-term engagement optimization reduced diversity and long-term retention, motivating a multi-objective ranking strategy balancing relevance, diversity, retention, latency, and scalability.
- Delivered a 14.3 percentage-point improvement in 30-day retention in A/B testing versus the NVIDIA Merlin production baseline.

### ETH Zürich

Zürich, Switzerland

PHD RESEARCHER

2020– 2024

- Introduced the first node-embedding framework connecting s–t min-cuts with contagion dynamics, improving node classification accuracy by up to 12%; also developed graph-Laplacian methods for link recommendation with provable guarantees.
- Designed the first scalable online algorithm for local clustering in multiplex networks, enabling efficient recovery of local structure without global graph exploration.
- Led SNSF-funded research from problem formulation through publication and mentored two Master's students on methodology, experiments, and interpretation.

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| <b>IIIT Hyderabad</b><br>HIGH PERFORMANCE COMPUTING RESEARCHER<br>• Developed high-performance parallel graph algorithms, achieving up to 4× speedups on large-scale graph problems. | Hyderabad, India<br>2013– 2016 |
| <b>NVIDIA</b><br>SENIOR SOFTWARE ENGINEER<br>• Developed operating system kernel-level drivers for NVIDIA Tegra systems deployed in BMW automotive platforms.                        | Bengaluru, India<br>2012– 2013 |

## Efficient LLM & Long-Range Information Flow Research

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| <b>Graph Cascades: Mesoscopic Rewiring Improves Sparse Graph Attention</b><br>Designed structured sparse-attention topologies for Graph Transformers that improve long-range information flow without dense attention, scaling to graphs with millions of nodes while improving predictive accuracy by up to 11%. | NeurIPS 2026 ORAL - First author         |
| <b>What Does Muon Amplify?</b><br>Characterized when spectral-reweighting optimizers such as Muon amplify useful rather than noise-dominated gradient directions during LLM pretraining, validating the analysis across Pythia models up to 1B parameters.                                                        | ICLR 2027 - First author - Under review  |
| <b>Schur Preconditioning for Long-Range Message Passing in GNNs</b><br>Applied preconditioning ideas from numerical linear algebra to mitigate over-squashing and improve long-range information flow in sparse neural architectures while approximately preserving the original interaction geometry.            | ICLR 2027 - Senior author - Under review |
| <b>KV-Cache Compression in LLMs by Elimination, Not Eviction</b><br>Recasts KV-cache compression as structured elimination that preserves interactions through removed tokens, retaining task-critical tokens in 88–100% of stress tests versus 12–50% for one-shot methods such as Top-K.                        | Manuscript 2026 - Sole author            |

## Graph Learning & Numerical Algorithms Research

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| <b>Canonical Schur Trees for Approximate Cholesky in Laplacian Systems</b><br>Developed randomized sparse Schur-tree preconditioners for Laplacian systems underlying large-scale graph learning systems achieving 4.4–4.7× speedups over state-of-the-art solvers. | KDD 2027 - First author - Under review     |
| <b>Thresholded Local Hyperflow Diffusion</b><br>Developed thresholded local hypergraph diffusion with exact local updates, improving clustering on noisy data while reducing active graph volume.                                                                   | AISTATS 2027 - First author - Under review |
| <b>Fast Local Graph Clustering under Noisy Data</b><br>Developed side-information-aware local graph clustering via shortest-path search, improving F1 by 6.4 points.                                                                                                | KDD 2027 - Senior author - Under review    |
| <b>Adjacency Search Embeddings</b><br>Developed node embeddings capturing higher-order structure, improving classification by up to 12.1% in GNNs.                                                                                                                  | TMLR 2025 - First author                   |

## Education

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| <b>ETH Zürich, Switzerland</b><br>PhD NETWORK SCIENCE<br>Dissertation: Influence Processes on Networks | 2020– 2024 |
| <b>Purdue University, USA</b><br>MSc COMPUTER SCIENCE — Focus: Numerical Optimization                  | 2016– 2019 |

## Academic Service & Recognition

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| <b>Awards:</b><br>Wallenberg AI Postdoctoral Fellowship   Best Paper Award, HiPC   Purdue–Qatar Fellowship   AICTE Fellowship |
| <b>Invited Talks:</b><br>Adjacency Search Embeddings, TU Wien, 2025   Influence Processes on Social Networks, UIUC, 2025      |
| <b>Academic Service:</b><br>Program Committee: AAAI   Peer Review: NeurIPS, ICML, AISTATS                                     |